Dr. Anika Sharma

Scientist E — Atmospheric Instrumentation & Applied AI

CONTACT

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TECHNICAL SKILLS

Avionics
MIL-STD-1553 / ARINC 429, Avionics integration

Scientific instrumentation
Environmental qualification testing, Sensor calibration, Radiosonde and upper-air systems

Embedded systems
Embedded C, ESP32 / STM32 microcontrollers

Edge AI
NVIDIA Jetson deployment

Drones & UAV systems
UAV payload integration

AI / machine learning
PyTorch, Deep learning

IoT
MQTT and Node-RED, InfluxDB and Grafana

EDUCATION

Ph.D., Electrical and Electronics Engineering
Northfield Institute of Science and Technology
2021 – ongoing
Thesis: Lightweight deep networks for real-time enhancement of aerial imagery

M.Tech., Avionics
Coastal Institute of Aeronautical Engineering
2010
Grade: CGPA 8.6/10
Thesis: Fault-tolerant data acquisition for airborne test instrumentation

B.E., Electronics and Communication Engineering
Northfield Institute of Science and Technology, Northfield University
2004
Grade: First Class with Distinction

CERTIFICATIONS & TRAINING

Deep Learning for Edge Devices

PROFESSIONAL SUMMARY

Electronics and avionics engineer with twelve years of airborne-systems integration and qualification testing, followed by ten years leading scientific instrumentation and edge-AI development in government service. Experienced in MIL-STD-1553 test rigs, UAV payload electronics, sensor telemetry, embedded GPU deployment and technical procurement.

KEY PROJECTS

Edge super-resolution pipeline for UAV imagery *(Ongoing)* Sep 2021 – Present
Principal investigator, Kestrel Atmospheric Instrumentation Laboratory
Research prototype that enhances low-resolution UAV video in real time on embedded GPUs using pruned and distilled deep networks.
Technologies: NVIDIA Jetson Xavier NX, NVIDIA Jetson Orin Nano, PyTorch, TensorRT, ONNX, Docker, Structured pruning, Knowledge distillation, INT8 quantisation

- Implemented structured channel pruning and knowledge distillation to shrink the super-resolution network for embedded deployment.
- Converted models through ONNX to INT8 TensorRT engines, achieving 3.1× faster inference on Jetson Xavier NX.

Radiosonde ground-station acceptance test protocol Jan 2018 – Mar 2019
Coordinator
Station acceptance testing varied between sites.

- A common acceptance protocol and checklist adopted for new ground-receiving systems.

IoT monitoring of laboratory infrastructure Feb 2020 – Jun 2022
Project lead
Instrument rooms lacked continuous environmental and power-quality monitoring.
Technologies: ESP32, Environmental sensors, MQTT, Node-RED, InfluxDB, Grafana

- Deployed ESP32 sensor nodes publishing over MQTT to a Node-RED flow, InfluxDB time-series storage and Grafana dashboards with threshold alerts.
- Continuous monitoring with automated alerts replaced manual twice-daily readings.

PROFESSIONAL EXPERIENCE

Scientist E Jul 2016 – Present
Upper Air Instrumentation Division, Kestrel Atmospheric Instrumentation Laboratory, New Delhi
Responsible for upper-air observation instruments, the division workshop and the laboratory's IoT and AI/ML activities.
Tools: Python, PyTorch, MQTT, Node-RED, InfluxDB, Grafana, Docker

- Lead a four-member AI/ML group developing lightweight super-resolution and detection models for UAV imagery, deployed on NVIDIA Jetson devices with TensorRT.
- Led acceptance testing and commissioning of radiosonde ground-receiving systems at three observing stations, writing the test protocol and the defect-tracking procedure.
- Designed and deployed an IoT monitoring network for laboratory infrastructure (ESP32 sensor nodes, MQTT, Node-RED, InfluxDB and Grafana dashboards) covering temperature, humidity and power quality.
- Prepared technical specifications and served on bid-evaluation committees for instrument procurement through the Government e-Marketplace in line with the General Financial Rules.
- Authored standard operating procedures for pre-flight checks and calibration of upper-air instruments used across the division.

Fictional sample profile — not a real person

Example Academy of Applied Computing
Mar 2022

Workshop on Efficient Deep Learning for Embedded Vision
Example Conference on Instrumentation (ECI 2023)
Dec 2023

National Seminar on Upper-Air Observing Systems
Example Society of Atmospheric Instrumentation
Aug 2021

Mid-Career Training Programme for Scientists
Example Institute of Scientific Administration (sponsored by Ministry of Example Sciences)
Feb 2020

Public Procurement and GFR 2017
Example Institute of Public Administration (sponsored by Kestrel Atmospheric Instrumentation Laboratory)
Nov 2019

Additional charge: **Workshop In-charge**, Upper Air Division Workshop — Apr 2019 – Present

Additional charge of the division workshop supporting instrument repair, fabrication and calibration.

- Introduced a job-card and spares-inventory system for instrument repairs and calibration work.
- Supervise technicians and plan preventive maintenance for workshop test equipment.

Nodal officer: **Nodal Officer, IoT and AI/ML Initiatives** — Jan 2021 – Present

Senior Design Engineer — Jun 2004 – Jun 2016

Avionics Integration Group, Halcyon Avionics Design Bureau, Bengaluru

Integration and qualification testing of airborne electronics and UAV payloads.

Tools: MIL-STD-1553, ARINC 429, LabVIEW, C, MATLAB

- Developed payload interface electronics for a fixed-wing UAV electro-optical sensor and supported its ground and flight trials.
- Integrated mission-computer and display units on MIL-STD-1553 avionics buses and built the automated test rig used for bench validation.
- Mentored graduate engineer trainees on avionics bus protocols and test documentation.

Graduate Engineer — Aug 2002 – May 2004

Test Systems Group, Meridian Electronics Limited, Bengaluru

Built automated test equipment for avionics line-replaceable units.

Tools: LabVIEW, C

- Developed LabVIEW test sequences for power-supply and interface boards of avionics units.

PUBLICATIONS

[1] **A. Sharma** and R. Iyer, "Knowledge-distilled lightweight networks for drone image super-resolution," accepted for publication in Example Transactions on Aerial Imaging.

[2] **A. Sharma**, K. Rao and R. Iyer, "Benchmarking embedded GPUs for real-time image enhancement," Example Symposium on Applied AI, pp. 88-93, Jan 2025.

[3] **A. Sharma** and R. Iyer, "Channel-pruned residual networks for real-time aerial image enhancement on embedded GPUs," Journal of Embedded Vision Systems, vol. 12, no. 3, pp. 45-58, Mar 2024. doi: 10.5555/jevs.2024.0123

[4] **A. Sharma**, P. Rao and S. Gill, "An IoT telemetry framework for upper-air ground stations," Proceedings of the Example Conference on Instrumentation (ECI 2023), pp. 112-117, Sep 2023.

[5] **A. Sharma** and S. Verma, "Edge telemetry dashboards for upper-air ground stations," Proceedings of the Example Conference on Instrumentation, Paper 17, Feb 2022.

[6] **A. Sharma**, P. Menon and R. Iyer, "A low-cost GNSS receiver chain for radiosonde tracking," Example Journal of Atmospheric Instrumentation, vol. 8, no. 2, pp. 101-112, Jun 2021. doi: 10.5555/ejai.2021.0042

MANUSCRIPTS UNDER REVIEW / IN PREPARATION

[1] **A. Sharma**, M. Khan and R. Iyer, "Detection-aware quality assessment of super-resolved drone images," under review at Example Transactions on Geoscience Imaging. *[Under review]*

[2] **A. Sharma**, "Simulation study of balloon-released gliding sondes," in preparation. *[In preparation]*

AWARDS & HONOURS

Best Paper Award — 2023

Example Conference on Instrumentation (ECI 2023)

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